

NEW FORMULAS FOR FALTINGS DELTA INVARIANT

1. MOTIVATION

Let $\mathbb{Q} \subseteq K$ be a number field of degree $d = [K : \mathbb{Q}]$ and C/K a smooth, projective curve of genus $g \geq 2$.

Theorem (Mordell conjecture/Faltings' theorem). *It holds $\#C(K) < \infty$.*

Open problem. Determine the set $C(K)$ effectively.

This was solved for curves over function fields by using intersection theory on the associated minimal fibering, which is a complete algebraic surface.

Arakelov's idea. Arakelov introduced an intersection theory for metrized line bundles on a model $p: \mathcal{C} \rightarrow B = \text{Spec } \mathcal{O}_K$ of C/K . One has canonical metrics for the dualizing sheaf $\omega_{\mathcal{C}/B}$ and $\det p_* \omega_{\mathcal{C}/B}$.

Theorem (Faltings–Noether formula). *Assume that C/K has semi-stable reduction over $\mathcal{O}_K$ and let $p: \mathcal{C} \rightarrow B$ be its minimal regular model. Then it holds*

$$12 \widehat{\deg} \det p_* \omega_{\mathcal{C}/B} = (\omega_{\mathcal{C}/B}, \omega_{\mathcal{C}/B}) + \sum_{v \in |B|} \delta_v \cdot \log \#k_v + \sum_{\sigma: K \rightarrow \mathbb{C}} \delta(C_\sigma) - 4g \log 2\pi,$$

where δ_v denotes the number of singularities on the geometric fibre at v and $\delta(C_\sigma)$ is Faltings delta invariant of the Riemann surface C_σ .

Parshin's method. If for all arithmetic surfaces $p: \mathcal{C} \rightarrow B$ of genus at least 2 there are constants c_1, c_2, c_3 depending only on the genus, such that

$$(\omega_{\mathcal{C}/B}, \omega_{\mathcal{C}/B}) \leq c_1 \left(\sum_{v \in |B|} \delta_v \cdot \log \#k_v + \sum_{\sigma: K \rightarrow \mathbb{C}} \delta(C_\sigma) \right) + c_2 \log |D_{K/\mathbb{Q}}| + c_3 d,$$

then an effective Mordell conjecture follows, even the *abc* conjecture.

2. FALTINGS DELTA INVARIANT

Definition by Faltings. Let X be a (compact and connected) Riemann surface of genus $g \geq 1$. Faltings defined $\delta(X)$ by

$$\|\theta\|(P_1 + \cdots + P_g - Q) = \exp\left(-\frac{1}{8}\delta(X)\right) \cdot \frac{\|\det(\omega_j(P_k))\|_{\text{Ar}}}{\prod_{j < k}^g G(P_j, P_k)} \cdot \prod_{j=1}^g G(P_j, Q),$$

where $\omega_1, \dots, \omega_g \in H^0(X, \Omega_X^1)$ is an orthonormal basis. Set $\mu = \frac{1}{2g} \sum_{j=1}^g \omega_j \wedge \overline{\omega_j}$.

The Arakelov–Green function is the unique function $G: X^2 \rightarrow \mathbb{R}_{\geq 0}$ satisfying (i) $\partial_P \overline{\partial}_P \log G(P, Q) = \pi i(\mu - \delta_Q)$ and (ii) $\int_X \log G(P, Q) \mu(P) = 0$. It defines a canonical metric on $\mathcal{O}_{\mathcal{C}}(P)$ by $\|1_{\mathcal{O}_{\mathcal{C}}(P)}\|_\sigma(Q) = G_{C_\sigma}(P, Q)$ for sections P, Q of p .

Let ν be the translation invariant $(1, 1)$ form on $\text{Pic}_0(X) \cong \text{Pic}_{g-1}(X)$ with $\int_{\text{Pic}_{g-1}(X)} \nu^g = g!$ and $\Theta \subseteq \text{Pic}_{g-1}(X)$ the divisor of line bundles having global sections. The function $\|\theta\|: \text{Pic}_{g-1}(X) \rightarrow \mathbb{R}_{\geq 0}$ is uniquely given by (i) $\partial \overline{\partial} \log \|\theta\| = \pi i(\nu - \delta_\Theta)$ and (ii) $\int_{\text{Pic}_{g-1}(X)} \|\theta\|^2 \nu^g = 2^{-g/2} g!$. It can explicitly be given by a theta series.

Kawazumi–Zhang invariant. Define h by $\partial_{X^2}\bar{\partial}_{X^2}\log G(P_1, P_2) = \pi i(h - \delta_\Delta)$ and the Kawazumi–Zhang invariant by $\varphi(X) = \int_{X^2} \log Gh^2$. It holds $\varphi \geq 0$. Further, define $H(X) = \int_{\text{Pic}_{g-1}(X)} \log \|\theta\|_{\frac{\nu_g}{g!}}$. We have $H(X) < -(g/4)\log 2$ by Jensen's inequality.

Theorem 1 (W.). *It holds $\delta(X) = -24H(X) + 2\varphi(X) - 8g\log 2\pi$.*

Strategy of proof. Consider $f(X) = 24H(X) + \delta(X) - 2\varphi(X)$ as function on $\mathcal{M}_g$. Show $\partial\bar{\partial}f(X) = 0$, which implies $f(X)$ is constant. Now proof $f(X) = -8g\log 2\pi$ if X is hyperelliptic using that the canonical involution on $\text{Pic}_{g-1}(X)$ factorizes by the hyperelliptic involution on X .

Corollary 1. *We have $\delta(X) > -2g\log 2\pi^4$.*

Let $\|\eta\|: \Theta \rightarrow \mathbb{R}_{\geq 0}$ be the derivative version of $\|\theta\|$ introduced by R. de Jong and define $\Lambda(X) = \int_{\Theta} \log \|\eta\|_{\frac{\nu_{g-1}}{g!}}$.

Theorem 2 (W.). *It holds $\delta(X) = 2(g-7)H(X) - 2\Lambda(X) - 4g\log 2\pi$. In particular, there is a canonical extension of δ to the moduli space of indecomposable PPAV's.*

The Arakelov–Green function. Bost proved that there exists an invariant $A(X)$ such that $\log G(P, Q) = \int_{\Theta+P-Q} \log \|\theta\|_{\frac{\nu_{g-1}}{g!}} + A(X)$.

Theorem 3 (W.). *It holds $A(X) = \frac{1}{2g}\varphi(X) - H(X)$.*

Corollary 2. *We have $\sup_{P, Q \in X} G(P, Q) < 2^{3g^3} \cdot \exp\left(\max\left(\frac{1}{4g}, \frac{2g+1}{48g}\right)\delta(X)\right)$.*

3. APPLICATIONS TO ARAKELOV THEORY

The Faltings height is given by $h_F(C) = \frac{1}{d}\widehat{\deg} \det p_*\omega_{\mathcal{C}/B}$. Let $P \in C(\bar{K})$. After a finite field extension of K , we can assume that $P \in C(K)$. The Arakelov height of P is given by $h(P) = \frac{1}{d}(\mathcal{O}_{\mathcal{C}}(P), \omega_{\mathcal{C}/B})$. Let $W \in C(\bar{K})$ be any Weierstraß point. By Corollary 1 and a formula by R. de Jong we obtain:

Corollary 3. *We have $h(W) < (6g^2 + 4g + 2)h_F(C) + 12g^4\log 2$.*

Szpiro's small points conjecture for cycle covers. If C is a cyclic cover of $\mathbb{P}_K^1$ of prime degree, there are infinitely many points $P \in C(\bar{K})$ satisfying

$$h(P) \leq r^{8g \, dr} (N_S \cdot |D_{K/\mathbb{Q}}|)^r - \min_{X \in \mathcal{M}_g} \delta(X) < r^{8g \, dr} (N_S \cdot |D_{K/\mathbb{Q}}|)^r + 2g\log 2\pi^4,$$

where $r = d(5g)^5$, S is the set of places of K , where C has bad reduction, and $N_S = \prod_{v \in S} \#k_v$. The first inequality is due to Javanpeykar and von Känel and the second inequality follows by Corollary 1.

Corollary 4 (Arithmetic Bogomolov–Miyaoko–Yau inequality for hyperelliptic curves). *If C is hyperelliptic, it holds*

$$(\omega_{\mathcal{C}/B}, \omega_{\mathcal{C}/B}) < \frac{g-1}{2g+1} \left((3g-1) \sum_{v \in |B|} \delta_v \cdot \log \#k_v + \sum_{\sigma: K \rightarrow \mathbb{C}} \delta(C_\sigma) + 2gd\log 2\pi^4 \right).$$

Remark. There are no arithmetic implications of Corollary 4 by Parshin's method.

Corollary 5 (Effective Bogomolov conjecture for hyperelliptic curves). *Assume C to be hyperelliptic. For any $z \in \text{Pic}_0(C)(\bar{K})$ there are only finitely many points $P \in C(\bar{K})$ satisfying*

$$h_{NT}(((2g-2)P - K_C) - z) \leq \frac{(g-1)^2}{(2g+1)d} \left(\frac{2g-5}{12g} \sum_{v \in |B|} \delta_v \cdot \log \#k_v + \sum_{\sigma: K \rightarrow \mathbb{C}} \varphi(C_\sigma) \right),$$

where K_C denotes the canonical divisor class of C .